Dibaloke Chanda

Teacher - Machine Learning Researcher

| dibaloke@eece.mist.ac.bd | Dibaloke-Chanda | Dibaloke-Chanda
| Dibaloke-Chanda | Dibaloke | 0000-0001-5993-659X | Personal Website

Work Experience

March 2021 - Present [Full Time]

Adjunct Lecturer
Military Institute of Science and Technology (MIST)

Course Taken

- Random Signal and Processes
- Electrical Drive and Instrumentation
- Electrical Circuit Analysis I
- Digital Signal Processing Laboratory
- Microwave Engineering Laboratory
- Numerical Techniques Laboratory
- Digital Communication Laboratory
- Electrical Drive and Instrumentation Laboratory
- Electronic Devices and Circuits Sessional
- Principle of Electrical Engineering Sessional

Mar 2021 - Present [Part Time]

Research Engineer
Visual Information Processing(VIP) Lab, MIST

- R&D on Computer Vision Applications, Machine Learning Architectures.
- Mentoring young researchers and conducting technical sessions.

Jul 2020 – Feb 2021

Senior Mentor (Android and IOS App Development)
MIST Innovation Club

Education

Feb 2017 - Feb 2021

Military Institute of Science and Technology, Bangladesh University of Professionals(Dhaka, Bangladesh)
B.Sc. in Electrical, Electronic and Communication Engineering
*(Major: **Communication**, CGPA: **3.98/4.00**)*

Research Interests

• Data Science • Deep Learning • Machine Learning • Computer Vision • Robotics • Human Computer Interaction

Relevant Courses

• Calculus • Vector analysis, Matrices and Geometry • Complex Variables and Statistics • Computer Programming • Continuous Signals and Linear Systems • Digital Communication • Digital Signal Processing • Random Signals and Processes

Technical Skills & Certifications

- **Programming Languages:** *Python, SQL, Matlab, C, LaTeX, Java, Javascript, PHP, Dart*
- **Deep Learning & Computer-vision Packages:** *Keras, TensorFlow, PyTorch, Scikit-Learn, Open-CV, Mediapipe*
- **Web Development & App Development:** *Nodejs, Django, Flask, Flutter*
- **Data Processing & Visualization:** *Numpy, Pandas, Matplotlib, Seaborn, Plotly*
- **Machine Learning:** *Time Series Prediction, Transfer Learning, Generative Adversarial Network, Convolutional Neural Network*
- **Robotics:** *Arduino, Raspberry Pi & Micro-controllers*
- **Graphics Design Software:** *Inkscape, Adobe XD*
- **Other Software:** *Pspice, Orcad, Proteus Design Suite*
- **Networking Software:** *Cisco Packet Tracer*
- **Selected Certifications:** *Computer Vision Basics, Data Science Math Skills, Managing Machine Learning Projects with Google Cloud, Understanding and Visualizing Data with Python, Inferential Statistical Analysis with Python, Deep Learning Specialization*

Test Scores

The International English Language Testing System (IELTS) - July 2022
Listening: 8.5 | Reading: 8.5 | Speaking: 6.5 | Writing: 6.5 | Overall: 7.5

Academic Research Experience

Feb 2020 – Feb 2021

An undergraduate thesis on Performance Analysis of Initialization Algorithms of Deep Neural Network Based Coordinated Beamforming System for mmWave
Supervisor: Brigadier General A K M Nazrul Islam, PhD, Dean FECE, Head of Department, Department of EECE, Military Institute of Science and Technology

Publications

2022

1. **Chanda, D.** & Roy, K. *A Novel Approach on Facial Emotion Recognition using Landmark Distance Based Deep Neural Network* in *International Conference on Information and Communication Technology for Development (ICICTD) [Under Review]* (2022).
2. Ovi, T. B., **Chanda, D.**, Naba, S. & Rizu, M. I. *A Novel Feature Extraction Based Double Ensemble Architecture For Improving Bangla Sign Language Recognition* in *International Conference on Information and Communication Technology for Development (ICICTD) [Under Review]* (2022).
3. Ovi, T. B., Naba, S. S., Onim, M. S. H. & **Chanda, D.** *A Transfer-Learning Based Ensemble Architecture for ECG Signal Classification* in *IEEE REGION 10 SYMPOSIUM [Accepted]* (2022).
4. Roy, K. & **Chanda, D.** *A Robust Webcam-based Eye Gaze Estimation System for Human-Computer Interaction* in *2022 International Conference on Innovations in Science, Engineering and Technology (ICISET)* (2022), 146–151.

2021

5. Alam, M. A., Hossain, S. M., **Chanda, D.** & Kabir, M. A. *Performance Analysis of LSTMs and Fbprophet Models for Short Term Load Forecasting* in *2021 5th International Conference on Electrical Engineering and Information & Communication Technology (ICEEICT)* (2021), 1–5.
6. **Chanda, D.** & Akash, M. I. *Performance Analysis Through Image and Video Transmission For Alamouti Space Time Block Coding Over Rayleigh And Rician Fading Channel* in *2021 5th International Conference on Electrical Engineering and Information & Communication Technology (ICEEICT)* (2021), 1–5.
7. **Chanda, D.**, Islam, A. N., *et al. Performance Analysis of Initialization Algorithms of Deep Neural Network Based Coordinated Beamforming System for mmWave* in *2021 5th International Conference on Electrical Engineering and Information & Communication Technology (ICEEICT)* (2021), 1–6.
8. Faiza, T. T., Haq, S. S., **Chanda, D.** & Halim, M. A. *A Study on the Performance Analysis of Hybrid Diversity Combining Techniques for Rayleigh and Rician Fading Channels under AWGN* in *2021 5th International Conference on Electrical Engineering and Information & Communication Technology (ICEEICT)* (2021), 1–6.
9. Halim, M. A., Haq, S. S., Faiza, T. T. & **Chanda, D.** *Comparative Analysis of Hybrid Diversity Schemes under AWGN and Impulsive Noise Models for Rayleigh Fading Channels* in *2021 5th International Conference on Electrical Engineering and Information & Communication Technology (ICEEICT)* (2021), 1–6.

Selected Projects & Research Works

Jan 2021 – Ongoing	**ECG Signal Classification with Ensemble of Deep Convolutional Network**
Dec 2021 – Ongoing	**Segmentation of Satellite Images with Generative Adversarial Network**
Dec 2021 – Ongoing	**Skin Cancer Classification with Ensemble Architecture**
Feb 2020 – Apr 2020	**Radar based industrial vehicular optimization and industrial safety enhancement**
Nov 2019 – Jan 2020	**Speaker Dependent Spoken Digit Recognation**
Jul 2019 – Sep 2019	**Intelligent irrigation system to improve agricultural output using IOT sensors**
Feb 2019 – May 2019	**Integrated home-security, automation & surveillance with Raspberry Pi, sensors & webcam**
Aug 2018 – Oct 2018	**FM Transmitter System**
Feb 2018 – May 2018	**Automatic Accident Detection and Emergency Response System**
Dec 2017 – Feb 2018	**Self-correcting dual axis solar panel for low income households**

Achievements and Other Credentials

2019 | **IEEE Conference Project Presentation: 1st Runner-up** | ***Awarded By:*** *ICEEICT*